

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Assignment 02



Submitted to:

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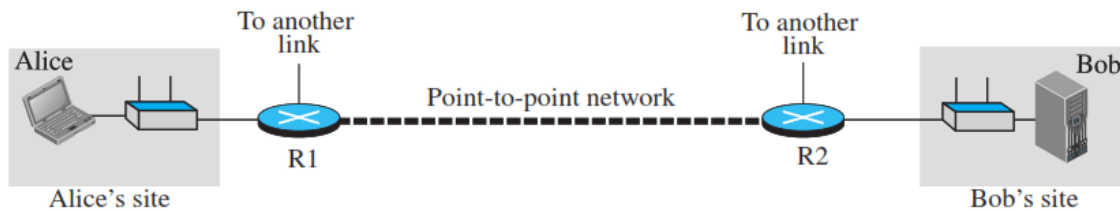
Subject: computer network

Department: BSSE-5-Afternoon

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In Figure 9.10, show the process of frame change in routers R1 and R2.

Figure 9.10 *The internet for our example*



Sender Alice:

At network layer a data packet is created with sender IP address and Bob's destination IP address.

At data link layer a frame is created with source Alice device's MAC address and destination R1 MAC address.

Router R1:

Data link layer at router R1 receive the frame and remove the header and create new one with source R1's MAC address and destination MAC address of R2.

Router R2;

Data link layer at router R2 receive the frame form source R1 and remove the header and create new one with source R2's MAC address and destination MAC address of Bob's device.

Bob's device:

Bob's device receive the frame at data link layer and remove the header and processes the IP packet at the network layer.

Question 2.

Assume Ali is travelling from 67, Main Street 3, E-11/4 in Islamabad to 1432, street 65, DHA Phase II in Karachi. He wants to travel by air from Islamabad to Karachi Airport.

BUT unfortunately, Ali cannot find a direct flight from the Islamabad to the Karachi.

If he needs to change flights in Lahore.

- a. Find the end-to-end addresses in this scenario.

The end-to-end address will remain constant. Source address will be "67, Main Street 3, E-11/4 Islamabad" and the destination address will be "1432, street 65, DHA Phase II in Karachi"

b. Find the link-layer addresses in this scenario.

Link layer address will be:

1. Ali home address "67, Main Street 3, E-11/4 Islamabad" to Islamabad airport.
2. Islamabad airport to Lahore airport.
3. Lahore airport to Karachi airport.
4. Karachi airport to the destination address "1432, street 65, DHA Phase II in Karachi"

Question 3.

Encode the data using Block coding technique of CRC, for transmission.

a. Check the data for any errors at the receiver side.

$$\text{Data} = x^9 + x^7 + x^6 + x^5 + x^3 + x + 1$$

$$\text{Div} = x^4 + x^3 + x^2$$

Answer:

transmitted codeword

DATE: ___/___/___

10) ①

Data = $x^9 + x^7 + x^6 + x^5 + x^3 + x + 1$
10011101011

Div = $x^4 + x^3 + x^2$
11100

11100 | 110111010110000

11100 ↓

10110

11100 ↓

10101

11100 ↓

10010

11100 ↓

11101

11100 ↓

11000

11100 ↓

1000

Transmitted codeword:
10111010111000

checking for error

DATE: ____/____/____

(a) 2

check for error

[illegible]

Remainder is zero so
no error

- b. Make an error in the 6th bit from MSB and again check for the errors.**

Making an error in the 6th bit from MSB and checking again.

$$\begin{array}{r}
 11100 \sqrt{101111 \square 10111000} \\
 \underline{11100} \downarrow \\
 10111 \\
 \underline{11100} \downarrow \\
 10111 \\
 \underline{11100} \downarrow \\
 10110 \\
 \underline{11100} \downarrow \\
 10101 \\
 \underline{11100} \downarrow \\
 10011 \\
 \underline{11100} \downarrow \\
 11111 \\
 \underline{11100} \downarrow \\
 11111 \\
 \underline{11100} \downarrow \\
 11000 \\
 \underline{11100} \\
 100
 \end{array}$$

Reminder is non-zero means there is an error.

- c. Consider the codeword about with error as an input for hamming.
Show how Hamming corrects the erroneous bit as entered above.

Codeword without error

1 0 1 1 1 0 1 0 1 1

1P 2P 3D 4P 5D 6D 7D 8P 9D 10D 11D 12D 13D 14D 15D

11P 17D 18P 19D

$P_1 = 1 \ 3 \ 5 \ 7 \ 9 \ 11 \ 13 \ 15 \ 17 \ 19$
1 0 1 1 1 1 1 1 = 1

$P_2 = 2 \ 3 \ 6 \ 7 \ 10 \ 11 \ 14 \ 15$
1 1 1 0 1 1 1 1 = 1

$P_4 = 4 \ 5 \ 6 \ 7 \ 12 \ 13 \ 14 \ 15$
0 1 1 0 1 1 1 1 = 0

$P_8 = 8 \ 9 \ 10 \ 11 \ 12 \ 13 \ 14 \ 15$
1 0 1 0 1 1 1 1 = 0

code word = 111001101011

with error: 11100110111011

$P_1 = 1 \ 3 \ 5 \ 7 \ 9 \ 11 \ 13 \ 15 \ 17 \ 19$
1 0 1 1 1 1 1 1 = 1 correct

$P_2 = 2 \ 3 \ 6 \ 7 \ 10 \ 11 \ 14 \ 15$
1 1 1 1 1 1 1 1 = 1 wrong

$P_4 = 4 \ 5 \ 6 \ 7 \ 12 \ 13 \ 14 \ 15$
0 1 1 0 1 1 1 1 = 0 correct

$P_8 = 8 \ 9 \ 10 \ 11 \ 12 \ 13 \ 14$
1 1 1 0 1 1 1 1 = 1 wrong

$2 + 8 = 10$

The 10th bit is an error so it must be change.